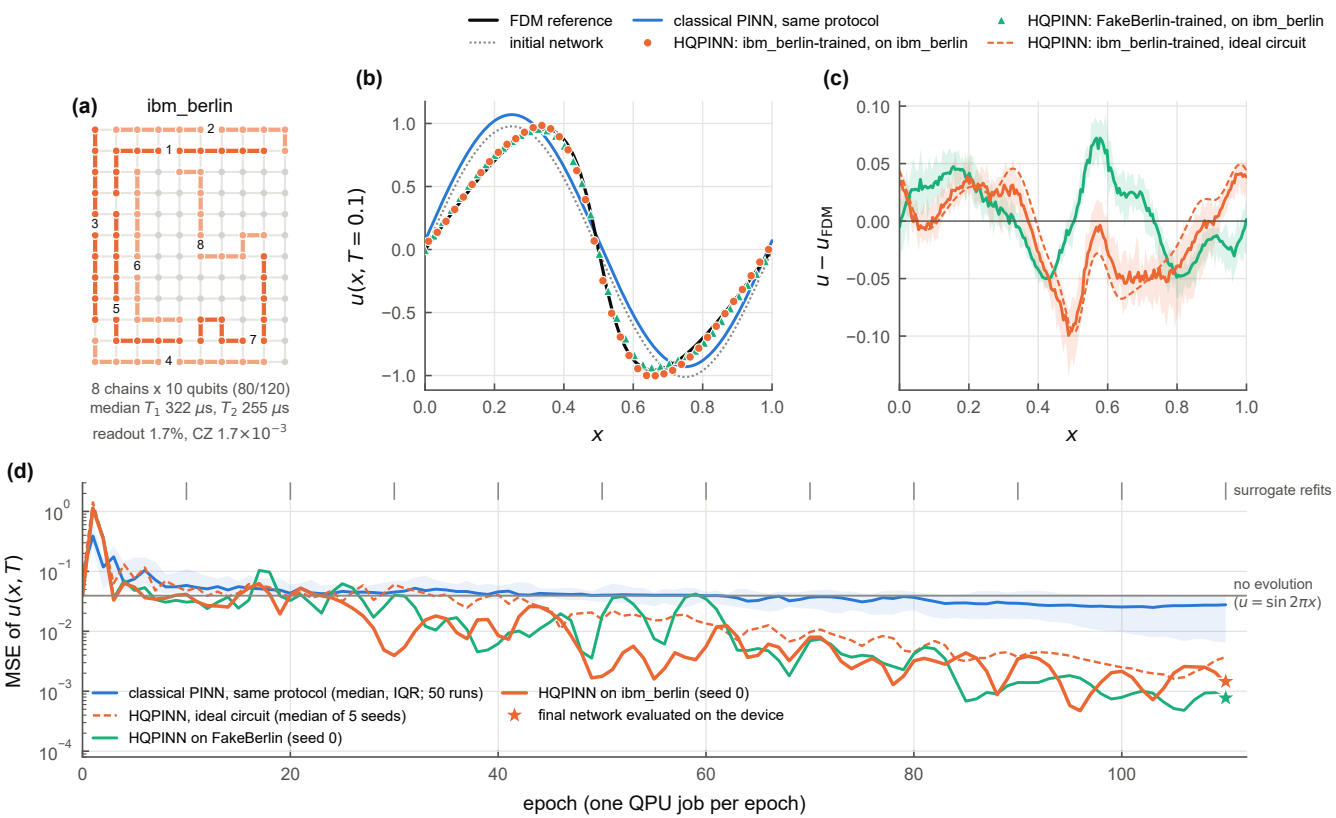

FDM reference
initial network
classical PINN, same protocol
HQPINN: ibm_berlin-trained, on ibm_berlin
HQPINN: FakeBerlin-trained, on ibm_berlin
HQPINN: ibm_berlin-trained, ideal circuit
(a)
ibm_berlin
8 chains x 10 qubits (80/120)
median T_1 322 μs, T_2 255 μs
readout 1.7%, CZ 1.7×10^{-3}
(b)
$u(x, T = 0.1)$
x
(c)
$u - u_{\mathrm{FDM}}$
x
(d)
MSE of $u(x, T)$
surrogate refits
no evolution
($u = \sin 2\pi x$)
classical PINN, same protocol (median, IQR; 50 runs)
HQPINN, ideal circuit (median of 5 seeds)
HQPINN on FakeBerlin (seed 0)
HQPINN on ibm_berlin (seed 0)
final network evaluated on the device
epoch (one QPU job per epoch)